Model Design of TCG-Net

1. Overall Architecture

We propose **TCG-Net**, a **Two-branch Cooperative Game-Theoretic Network** for individualized cognitive assessment using functional connectivity and plasma proteomics.

The model takes two types of inputs:

$$X_p \in \mathbb{R}^{N \times P}$$
$$A_f \in \mathbb{R}^{N \times R \times R}$$

where $X_p$denotes the plasma proteomic feature matrix, $P = 2920$is the number of proteins, $A_f$denotes the individualized functional connectivity matrix derived from resting-state fMRI, $R$is the number of brain regions, and $N$is the number of participants.

The model consists of four major components:

1. **Proteomics Encoder**
2. **fMRI Connectome Encoder**
3. **Cooperative Shapley Fusion Module**
4. **Prediction and Interpretation Head**

The overall pipeline is:

$$X_p \rightarrow \text{Proteomics Encoder} \rightarrow z_p$$
$$A_f \rightarrow \text{fMRI Connectome Encoder} \rightarrow z_f$$
$$(z_p, z_f) \rightarrow \text{Cooperative Shapley Fusion Module} \rightarrow z_{fusion}$$
$$z_{fusion} \rightarrow \text{Prediction Head} \rightarrow \hat{y}$$

where $z_p$and $z_f$represent modality-specific latent embeddings, and $\hat{y}$denotes the predicted cognitive score.

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2. Proteomics Encoder

The proteomics branch is designed to extract biologically meaningful latent representations from high-dimensional plasma proteomic profiles. Since the current proteomics pipeline has already been validated using UK Biobank data, this branch can serve as a stable foundation for multimodal integration.

2.1 Input

The input to the proteomics branch is:

$$X_p \in \mathbb{R}^{B \times 2920}$$

where $B$is the batch size.

Before being fed into the model, the protein features should be standardized using z-score normalization or covariate-residualized and standardized values.

2.2 Network Structure

The proteomics encoder is implemented as a **Squeeze-and-Excitation Multi-Layer Perceptron**, or **SE-MLP**.

Recommended architecture:

```
Input: 2920-dimensional protein vector

Linear(2920, 1024)
BatchNorm1d(1024)
GELU
Dropout(0.3)

SE-Block(1024)

Linear(1024, 512)
BatchNorm1d(512)
GELU
Dropout(0.3)

Linear(512, 256)
BatchNorm1d(256)
GELU
Dropout(0.2)

Linear(256, 128)

Output: z_p, dimension = 128
```

The final proteomic embedding is:

$$z_p \in \mathbb{R}^{B \times 128}$$

2.3 Squeeze-and-Excitation Block

The SE-block is used to learn non-linear protein-channel importance weights. Given an intermediate protein representation:

$$h_p \in \mathbb{R}^{B\times 1024}$$

the SE-block computes:

$$s_p = \sigma(W_2\delta(W_1 h_p))$$

where:

- $W_1: 1024 \to 256$
- $W_2: 256 \to 1024$
- $\delta$is the GELU activation function
- $\sigma$is the sigmoid activation function

The reweighted protein representation is:

$$\tilde{h}_p = h_p \odot s_p$$

where $s_p$can be interpreted as a protein-channel importance vector.

A simplified PyTorch-style implementation is:

```
class SEBlock(nn.Module):
    def __init__(self, dim, reduction=4):
        super().__init__()
        self.fc = nn.Sequential(
            nn.Linear(dim, dim // reduction),
            nn.GELU(),
            nn.Linear(dim // reduction, dim),
            nn.Sigmoid()
        )

    def forward(self, x):
        weight = self.fc(x)
        return x * weight, weight
```

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3. fMRI Connectome Encoder

The fMRI branch is designed to encode individualized functional brain networks. Rather than simply vectorizing the functional connectivity matrix, we recommend using a **connectome-aware graph neural network**.

3.1 Input Functional Connectivity Matrix

For each participant, the fMRI input is an individualized functional connectivity matrix:

$$A_i \in \mathbb{R}^{R \times R}$$

where Rdenotes the number of brain regions.

Each element $A_i(r, s)$represents the functional connectivity strength between brain regions $r$and s, usually computed using Pearson correlation followed by Fisher z-transformation.

3.2 Node Feature Construction

For each brain region, we recommend constructing node features by combining two types of information.

3.2.1 FC Profile Features

For each brain region r, its connectivity profile is defined as:

$$x_i^{FC}(r) = A_i[r, :]$$

Thus, for each participant:

$$X_i^{FC} \in \mathbb{R}^{R \times R}$$

This representation preserves the whole-brain connectivity profile of each node.

3.2.2 Graph-Theoretical Features

Additional graph-theoretical nodal metrics can be computed for each brain region, including:

1. Nodal strength
2. Clustering coefficient

3. Betweenness centrality
4. Participation coefficient
5. Local efficiency
6. Within-module degree

This gives:

$$X_i^{graph} \in \mathbb{R}^{R \times G}$$

where $G$is the number of graph-theoretical features.

The final node feature matrix is:

$$X_i = [X_i^{FC}, X_i^{graph}]$$
$$X_i \in \mathbb{R}^{R \times (R+G)}$$

For example, if $R = 66$and $G = 6$, the node feature dimension is 72.

3.3 Graph Edge Construction

We recommend using a **group-level sparse mask with individual edge weights** as the primary graph construction strategy.

First, a group-level functional connectivity mask is constructed using only the training set:

$$M_{group} \in \{0,1\}^{R \times R}$$

For example, the top 20% strongest group-level absolute FC edges are retained.

Then, for each individual:

$$A_i^{masked} = A_i \odot M_{group}$$

This strategy has several advantages:

- It preserves individual-level FC weights.
- It reduces noise from weak or unstable edges.
- It avoids information leakage by constructing the mask only from the training set.
- It provides a stable topology for graph neural network learning.

As sensitivity analyses, two additional graph definitions can be tested:

Alternative 1: Fully weighted individual FC graph

$$A_i = FC_i$$

Alternative 2: Positive and negative FC dual-channel graph

$$A_i^+ = \max(A_i, 0)$$
$$A_i^- = |\min(A_i, 0)|$$

The positive and negative FC matrices can be encoded by two separate GNN encoders and then concatenated.

3.4 GNN Architecture

The recommended fMRI encoder is a hybrid **GCN-GAT connectome encoder**.

```
Input:
Node features X: R × D
Adjacency matrix A: R × R

GraphConv(D, 128)
BatchNorm
GELU
Dropout(0.2)

GraphAttention(128, 128, heads=4)
BatchNorm
GELU
Dropout(0.2)

GraphConv(128 × 4, 128)
GELU

Global Mean Pooling
Global Max Pooling
Concatenate

Linear(256, 128)

Output: z_f, dimension = 128
```

The output fMRI embedding is:

$$z_f \in \mathbb{R}^{B\times 128}$$

The GAT layer also provides attention coefficients that can later be used for node-level and edge-level interpretation.

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4. Cooperative Shapley Fusion Module

The **Cooperative Shapley Fusion Module** is the key methodological innovation of TCG-Net. Instead of simply concatenating latent embeddings or learning unconstrained attention weights, TCG-Net explicitly estimates the contribution of each modality and their interaction using a cooperative game-theoretic formulation.

4.1 Three Prediction Pathways

To estimate modality-specific contributions, the model generates three auxiliary predictions.

fMRI-only prediction

$$\hat{y}_f = H_f(z_f)$$

Proteomics-only prediction

$$\hat{y}_p = H_p(z_p)$$

Joint prediction

$$\hat{y}_{fp} = H_{fp}(z_{joint})$$

where the joint representation is defined as:

$$z_{joint} = [z_f, z_p, z_f \odot z_p, | z_f - z_p |]$$

Since z_f and z_p are both 128-dimensional, the joint representation has dimension:

$$128 + 128 + 128 + 128 = 512$$

This design explicitly captures:

- fMRI-specific information
- proteomics-specific information
- multiplicative cross-modal interactions
- absolute cross-modal differences

4.2 Individual-Level Shapley Contribution Estimation

For two modalities, Shapley values can be computed explicitly.

Let $\hat{y}_0$be the baseline prediction, defined as the mean cognitive score in the training set:

$$\hat{y}_0 = mean(y_{train})$$

The fMRI contribution is computed as:

$$\phi_f = \frac{1}{2}[(\hat{y}_f - \hat{y}_0) + (\hat{y}_{fp} - \hat{y}_p)]$$

The proteomics contribution is computed as:

$$\phi_p = \frac{1}{2}[(\hat{y}_p - \hat{y}_0) + (\hat{y}_{fp} - \hat{y}_f)]$$

The interaction contribution is defined as:

$$\phi_{int} = \hat{y}_{fp} - \hat{y}_0 - \phi_f - \phi_p$$

Here:

- $\phi_f$represents the individual-level contribution of the functional connectome.
- $\phi_p$represents the individual-level contribution of plasma proteomics.
- ϕ_{int}represents the interaction contribution between fMRI and proteomics.

4.3 Shapley-Guided Fusion Gate

The Shapley contributions are transformed into fusion weights:

$$[w_f, w_p, w_{int}] = softmax([\phi_f, \phi_p, \phi_{int}])$$

Then three modality-specific latent components are constructed:

$$z'_f = W_f z_f$$
$$z'_p = W_p z_p$$
$$z'_{int} = W_{int}(z_f \odot z_p)$$

The final fused representation is:

$$z_{fusion} = w_f z'_f + w_p z'_p + w_{int} z'_{int}$$

where:

$$z_{fusion} \in \mathbb{R}^{B \times 128}$$

This is the core of TCG-Net: the Shapley contributions are not only used for post-hoc explanation but also directly guide multimodal latent fusion.

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5. Prediction Head

The final prediction head maps the fused representation to the cognitive outcome.

```
Input: z_fusion, 128

Linear(128, 64)
GELU
Dropout(0.2)

Linear(64, 32)
GELU

Linear(32, 1)

Output: predicted cognitive score
```

For single-task prediction:

$$\hat{y} \in \mathbb{R}^{B \times 1}$$

For multi-task cognitive prediction:

$$\hat{Y} \in \mathbb{R}^{B \times K}$$

where K denotes the number of cognitive outcomes.

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6. Loss Function

The total training objective is:

$$\mathcal{L} = \mathcal{L}_{main} + \lambda_1 \mathcal{L}_{aux} + \lambda_2 \mathcal{L}_{shapley} + \lambda_3 \mathcal{L}_{sparse}$$

6.1 Main Prediction Loss

For continuous cognitive scores, the primary loss can be mean squared error:

$$\mathcal{L}_{main} = MSE(y, \hat{y})$$

Alternatively, Huber loss can be used to improve robustness to outliers:

$$\mathcal{L}_{main} = Huber(y, \hat{y})$$

For UK Biobank cognitive phenotypes, Huber loss may be preferable because cognitive measures can contain extreme values and non-Gaussian distributions.

6.2 Auxiliary Prediction Loss

Auxiliary losses are used to ensure that both modality-specific encoders learn meaningful predictive representations:

$$\mathcal{L}_{aux} = MSE(y, \hat{y}_f) + MSE(y, \hat{y}_p) + MSE(y, \hat{y}_{fp})$$

This prevents one branch from collapsing during joint training.

6.3 Shapley Efficiency Loss

The Shapley decomposition should satisfy the efficiency property:

$$\hat{y}_{fp} \approx \hat{y}_0 + \phi_f + \phi_p + \phi_{int}$$

Therefore:

$$\mathcal{L}_{shapley} = \| \hat{y}_{fp} - (\hat{y}_0 + \phi_f + \phi_p + \phi_{int}) \|^2$$

If ϕ_{int} is defined exactly as above, this loss will be close to zero by construction. It can still be retained as a numerical stability constraint.

6.4 Sparsity Loss

To improve interpretability, sparse regularization can be applied to modality weights, protein-channel weights, and graph attention coefficients:

$$\mathcal{L}_{sparse} = || w ||_1 + || s_p ||_1 + || \alpha_f ||_1$$

where:

- w denotes modality fusion weights.
- s_p denotes SE-block protein-channel weights.
- α_f denotes GAT attention coefficients or edge-level importance scores.

Recommended initial hyperparameters:

```
lambda_aux = 0.3
lambda_shapley = 0.1
lambda_sparse = 1e-4
```

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7. Training Strategy

7.1 Stage 1: Single-Modality Pretraining

First, train each modality branch independently.

Proteomics pretraining

$$X_p \rightarrow z_p \rightarrow \hat{y}_p$$

fMRI pretraining

$$A_f \rightarrow z_f \rightarrow \hat{y}_f$$

The purpose of single-modality pretraining is to:

1. Ensure that each encoder learns predictive representations.
2. Prevent one modality from dominating the fusion process.
3. Provide stable initialization for joint training.

7.2 Stage 2: Joint TCG-Net Training

After pretraining, the two encoders are loaded into the full TCG-Net model. The joint prediction head, Shapley contribution module, Shapley-guided fusion gate, and final prediction head are then trained.

Recommended training protocol:

```
First 10 epochs:
Freeze both encoders.
Train only the fusion module and prediction heads.

After 10 epochs:
Unfreeze all modules.
Fine-tune the entire network using a lower learning rate.
```

Recommended learning rates:

```
Stage 1 pretraining:
lr = 1e-3

Stage 2 frozen fusion training:
lr = 5e-4

Stage 2 fine-tuning:
```

```
encoder lr = 1e-5
fusion/head lr = 1e-4
```

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8. Model Variants

To support systematic experiments and ablation analyses, we recommend implementing three versions of TCG-Net.

8.1 TCG-Net-Simple

This is the minimal working version.

```
Proteomics Encoder: SE-MLP
fMRI Encoder: vectorized FC + MLP
Fusion: Shapley-guided gate
Output: cognitive score
```

Purpose:

- Test whether Shapley-guided fusion works.
- Verify whether bimodal fusion improves prediction.
- Generate preliminary modality-contribution results.

8.2 TCG-Net-GNN

This is the recommended main model.

```
Proteomics Encoder: SE-MLP
fMRI Encoder: GCN-GAT connectome encoder
Fusion: Shapley-guided cooperative fusion
Output: cognitive score
```

Purpose:

- Preserve brain network topology.
- Improve connectome-level representation learning.
- Provide brain-region and network-level interpretability.

8.3 TCG-Net-Interaction

This is an extended model.

```
Proteomics Encoder: SE-MLP
fMRI Encoder: positive/negative FC dual-GNN
Fusion: modality Shapley + interaction Shapley
Output: multi-task cognitive prediction
```

Purpose:

- Separate positive and negative functional connectivity.
- Model stronger cross-modal interactions.
- Support multi-domain cognitive prediction.

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9. Recommended Main Model

The final recommended model is:

```
Input 1:
2920 plasma proteins

Proteomics Encoder:
Linear 2920 → 1024
BatchNorm + GELU + Dropout
SE-Block
Linear 1024 → 512
BatchNorm + GELU + Dropout
Linear 512 → 256
BatchNorm + GELU
Linear 256 → 128
Output: z_p

Input 2:
R × R functional connectivity matrix

fMRI Encoder:
Node features = FC profile + graph-theoretical metrics
Adjacency = group-level sparse mask × individual FC weights
GCN layer
GAT layer
GCN layer
Global mean pooling + global max pooling
Linear 256 → 128
Output: z_f

Auxiliary Heads:
z_p → y_p
z_f → y_f
[z_p, z_f, z_p × z_f, |z_p - z_f|] → y_fp

Shapley Module:
```

```
Compute phi_p, phi_f, phi_int
Apply softmax to obtain w_p, w_f, w_int

Fusion:
z_fusion = w_p W_p z_p + w_f W_f z_f + w_int W_int(z_p × z_f)

Prediction Head:
Linear 128 → 64
GELU + Dropout
Linear 64 → 32
GELU
Linear 32 → 1 or K

Outputs:
Predicted cognition
Individual-level modality contribution
Protein importance
Brain-region and edge importance
```

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10. Ablation Models

The following ablation models should be implemented to demonstrate the necessity of each component.

Model	Design	Purpose
M1 Proteomics-only	SE-MLP	Protein baseline
M2 fMRI-only	GNN	Brain connectome baseline
M3 Early fusion	Raw or latent feature concatenation	Conventional fusion baseline
M4 Late fusion	Weighted average of $\hat{y}_p$ and $\hat{y}_f$	Prediction-level fusion
M5 Attention fusion	Learned attention weights	Standard deep fusion
M6 Gated fusion	Learned modality gate	Non-Shapley gate baseline
M7 TCG-Net without interaction	Only ϕ_f and ϕ_p	Test interaction term
M8 TCG-Net without SE	Replace SE-MLP with ordinary MLP	Test SE-block
M9 TCG-Net without GNN	Replace GNN with vectorized FC MLP	Test graph modeling
M10 Full TCG-Net	Complete model	Main model

The expected performance pattern is:

$$M10 > M6 > M5/M4 > M2 > M1$$

However, even if the full model shows only modest predictive improvement over standard gated fusion, its individualized interpretability can still provide substantial scientific value.

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11. Model Outputs for Interpretation

The final model should output more than a predicted cognitive score. It should generate the following interpretable quantities:

1. Final cognitive prediction: $\hat{y}$
2. Proteomics-only prediction: $\hat{y}_p$
3. fMRI-only prediction: $\hat{y}_f$
4. Joint prediction: $\hat{y}_{fp}$
5. Individual-level modality contributions: $\phi_p, \phi_f, \phi_{int}$
6. Protein importance scores from SE weights, gradient attribution, or perturbation analysis
7. Brain-region and edge importance scores from GAT attention, integrated gradients, or edge perturbation analysis

This allows TCG-Net to function not only as a prediction model but also as an individualized mechanism-discovery framework.

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12. Individualized Mechanistic Subtyping

Based on Shapley contributions, participants can be categorized into three mechanistic subtypes.

Define:

$$D_i = \phi_f^i - \phi_p^i$$

Then standardize:

$$D_i^z = \frac{D_i - mean(D)}{std(D)}$$

Subtyping rule:

```
D_z > 0.5      brain-dominant group
D_z < -0.5     blood-dominant group
otherwise      balanced group
```

The three groups can then be compared in terms of:

1. Age
2. Cognitive performance
3. Reaction time
4. Vascular risk
5. Inflammatory burden
6. Biological age gap
7. Brain network efficiency
8. Mental health phenotypes

This analysis can reveal whether cognitive impairment is more strongly associated with brain network dysfunction, peripheral molecular dysregulation, or combined cross-scale mechanisms in different individuals.

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13. Main Innovation Statement

A concise methodological innovation statement could be:

Unlike conventional multimodal fusion models that concatenate latent representations or learn unconstrained attention weights, TCG-Net introduces a cooperative game-theoretic fusion mechanism in which modality-specific and interaction-level Shapley contributions are explicitly estimated for each individual and used to guide latent fusion. This design enables the model to simultaneously improve cognitive prediction and provide individualized mechanistic interpretation of brain-dominant, blood-dominant, and cross-scale interaction patterns.

A more paper-style version:

TCG-Net provides an interpretable cross-scale fusion framework for cognitive assessment by integrating functional connectomes and plasma proteomics through a cooperative game-theoretic mechanism. By embedding Shapley contribution estimation into the fusion layer, the model not only learns predictive multimodal representations but also quantifies how brain network and peripheral molecular information contribute to cognitive variation at the individual level.

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14. Minimal Viable Implementation

To reduce implementation risk, we recommend first implementing a minimal version before moving to the full GNN model.

MVP version

```
Proteomics branch:
SE-MLP → z_p

fMRI branch:
Vectorized FC + MLP → z_f

Auxiliary predictions:
z_p → y_p
z_f → y_f
[z_p, z_f, z_p × z_f, |z_p - z_f|] → y_fp

Shapley module:
Compute phi_p and phi_f
Apply softmax to obtain modality weights

Fusion:
weighted fusion of z_p and z_f

Prediction:
final cognitive score
```

The MVP should first demonstrate:

1. The Shapley-guided fusion module runs stably.
2. Bimodal fusion outperforms single-modality prediction.
3. Individual-level modality contributions show meaningful biological heterogeneity.

After the MVP is validated, the fMRI MLP branch can be replaced with the full GCN-GAT connectome encoder.